b) Dining-Philosopher's problem

```
#include <stdio.h>

#define N 5

void dining(int ph)
{
   printf("\nPhilosopher %d is Thinking", ph);

   printf("\nPhilosopher %d takes Fork %d and Fork %d",
         ph, ph, (ph + 1) % N);

   printf("\nPhilosopher %d is Eating", ph);

   printf("\nPhilosopher %d puts down Fork %d and Fork %d\n",
         ph, ph, (ph + 1) % N);
}

int main()
{
   int ph;

   printf("Enter Philosopher Number (0-4): ");
   scanf("%d", &ph);

   if(ph >= 0 && ph < N)
      dining(ph);
   else
      printf("Invalid Philosopher Number");

   return 0;
}
```

OUTPUT:

```
Enter Philosopher Number (0-4): 4

Philosopher 4 is Thinking
Philosopher 4 takes Fork 4 and Fork 0
Philosopher 4 is Eating
Philosopher 4 puts down Fork 4 and Fork 0

Process returned 0 (0x0)   execution time : 1.567 s
Press any key to continue.
```